PROGRAM 6: Text-to-Speech for Engineering Text

AIM

To convert engineering-related text into speech using a pretrained text-to-speech model.

ALGORITHM

1. Import the required libraries.
2. Load a pretrained text-to-speech model.
3. Accept engineering-related text.
4. Convert the text into audio.
5. Play and save the generated audio.

CODE

```
from transformers import pipeline
from IPython.display import Audio, display
import scipy.io.wavfile as wavfile

tts = pipeline(
    "text-to-speech",
    model="facebook/mms-tts-eng"
)

text = input("Enter engineering text: ")

result = tts(text)

audio = result["audio"]
sample_rate = result["sampling_rate"]

display(Audio(audio.squeeze(), rate=sample_rate))

wavfile.write(
    "engineering_speech.wav",
```

```
    sample_rate,
    audio.squeeze()
)

print("Audio generated successfully.")
```

INPUT

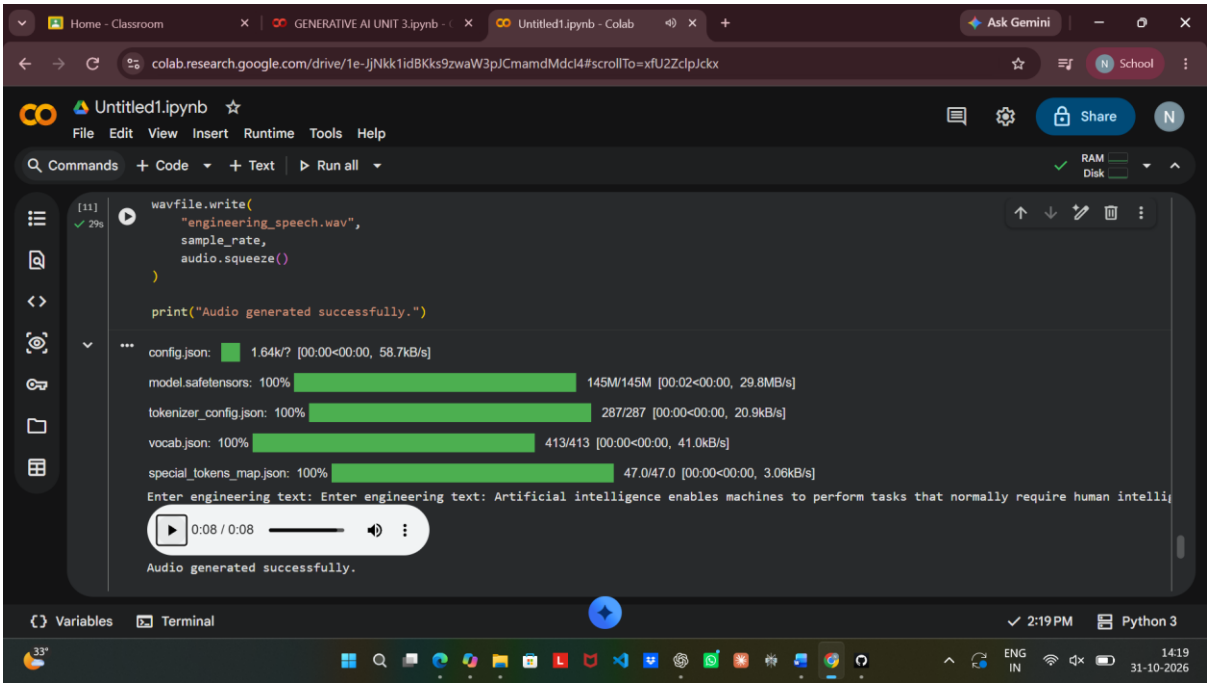
Enter engineering text:

Artificial intelligence enables machines to perform tasks

that normally require human intelligence.

OUTPUT

- The computer generates and plays the spoken audio.
- An audio file named engineering_speech.wav is saved.



RESULT

Thus, engineering-related text was successfully converted into speech using a pretrained text-to-speech model.